

Arda Kayaalp

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PERSONAL INFORMATION

Birth: July 8, 1997, Turkey

Gender: Male

Citizenship: Turkish

EDUCATION

Middle East Technical University

Mar. 2021 – Aug. 2024

M.Sc. in Physics, Department of Physics (3.36/4.00 CGPA)

Ankara, Turkey

Thesis Title: Application of the Quantal Diffusion Approach Based on the Stochastic Mean-Field Theory

Hacettepe University

Aug. 2015 – Jun. 2019

B.Sc. in Nuclear Engineering (Honours, 3.22/4.00 CGPA)

Ankara, Turkey

Graduation Project : A Study of $p - B^{11}$ Fusion Reaction using Geant4 Toolkit

EMPLOYMENT & RESEARCH EXPERIENCE

R&D Engineer

Jul. 2022 – May 2023

Oncotech Medical Systems

Ankara, Turkey

- Research and algorithm development on medical data and image processing systems.
- Development of MATLAB-based prototype of a DICOM-RT viewer application with GUI.
- Research and algorithm development on a low-cost, fast and accurate optical surface tracking system for SGRT applications.
- Development of a low-cost prototype application of an SGRT system in C++ using OpenCV and CUDA toolkits.

Scientific Project Specialist

Jan. 2020 – Jun. 2022

Research and Application Center for Space and Accelerator Technologies at METU

Ankara, Turkey

- Magnetic alignment calculations of quadrupole magnets on METU-DBL.
- Proton beam parameter calculations and beam preparations for the irradiation tests.
- Periodic maintenance of the beamline elements on the METU-DBL.
- Monte Carlo simulations and radiation dose calculations for the irradiation tests.
- Research and analysis on chaotic and quantum random systems.
- Development of image processing applications to extract information from chaotic systems.
- Mentored and tutored interns, reviewed their progress, and provided guidance to ensure their development and successful integration into the team.
- Compiled and presented progress reports, summarizing project developments and outcomes for senior management.

Undergraduate Researcher

Dec. 2018 – Jan. 2020

Research and Application Center for Space and Accelerator Technologies at METU

Ankara, Turkey

- Developed MATLAB-based image processing software to extract beam diagnostic data from the beam screen.
- Design and implementation of irradiation tests on METU-DBL.
- Basic proton beamline calculations with TURTLE Program.

**METU-DBL : Middle East Technical University Defocusing Beam Line*

RESEARCH BACKGROUND

Multinucleon transfer mechanisms, computational radiation transport, radiation detection and measurement.

INTERNSHIPS

Çekmece Nuclear Research and Education Center

Studied reactor physics, radiation protection and radioactive waste management.

Jun. – Jul. 2018

Istanbul, Turkey

OMV Samsun Thermal Power Plant

Studied thermodynamical aspects of a power plant.

Jul. – Aug. 2017

Samsun, Turkey

AWARDS & GRANTS

The Scientific and Technical Research Council of Turkey (TUBITAK)

TUBITAK Graduate Scholarship

Mar. 2022 – Aug. 2024

Ankara, Turkey

Random Power Hackathon, organised by randompower.eu & CERN Openlab

Third Place as Team Hexagon

Oct. 2020

Virtual Event

Project : Random Power generated TRNs for Geant4 HEP event simulations

TECHNICAL SKILLS

Hardware Experience : I possess fundamental skills in electronic laboratory work and soldering, along with hands-on experience in radiation detector electronics and radiation counting statistics. Furthermore, I have a basic understanding of ARM-based Linux development and proficiency in edge computing, making use of platforms such as Jetson Nano and Raspberry Pi.

Software Experience :

- **MATLAB:** Moderate to advanced proficiency. Developed radiation shielding models, detector readout interfaces, and applications for data and image processing. Created several GUI-based prototype applications.
- **Python:** Intermediate proficiency. Utilized for data processing, analysis, and visualization in various projects.
- **C++:** Intermediate proficiency. Employed in developing Geant4-based particle simulations and currently used for nuclear evaporation calculations.
- **Fortran:** Basic proficiency. Applied for simple manipulations of legacy physics codes.
- **CMAKE:** Basic proficiency. Used to build and manage cross-platform C++ applications.
- **Git:** Basic proficiency. Utilized for version control in local development projects.
- **MCNP & SERPENT:** Basic to intermediate proficiency. Modeled radiation shielding geometries with MCNP and calculated reactivity and modeled fuel burn-up cycles for LWR fuel rod geometry using SERPENT.
- **Geant4:** Basic to intermediate proficiency. Modeled simple nuclear interactions and radiation transport problems.
- **Linux:** Intermediate proficiency. Utilized as the primary development OS for all projects.
- **LaTeX:** Intermediate proficiency. Used for writing academic papers and various types of documentation.

LANGUAGE PROFICIENCY

- Turkish - Native
- English - TOEFL iBT Score : 104 (C1)

TEACHING EXPERIENCE

I am occasionally tutoring undergraduate students on applications of nuclear physics courses.

EXTRACURRICULAR ACTIVITIES

Digital photography, charcoal drawing, fountain pen collecting.

PUBLISHED WORKS

1. Kayaalp, A. *et al.* A quantal diffusion approach for multinucleon transfer in heavy-ion collisions. *Nuclear Physics A*, 122916 (2024)
2. Kayaalp, A. *et al.* A theoretical study on quasifission and fusion–fission processes in heavy-ion collisions. *The European Physical Journal A* **60**, 79 (2024)

REFERENCES

- **Prof. Osman Yılmaz - M.Sc. Academic Advisor**
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- **Prof. Şakir Ayık - M.Sc. Academic Co-Advisor**
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